UNIVERSITI MALAYA

NFMN

SIM 1005 / SIN 1002

Fundamentals of Spreadsheets / Introduction to Worksheet

Lecture 8: Introduction to VBA

By

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Objectives

- To define VBA, its components and tools in Excel.
- To create Macro in Excel using VBA.
- To execute CommandButton in Excel.
- To explore Visual Basic Editor and range of object.
- To manipulate types of variable in VBA project.

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VBA

- Visual Basic for Applications is a programming language in Excel
- VBA includes:
  1) Creating Macro (to automate tasks)
  2) MSGBox
  3) Workbook/Worksheet Object
  4) Range Object
  5) Variables

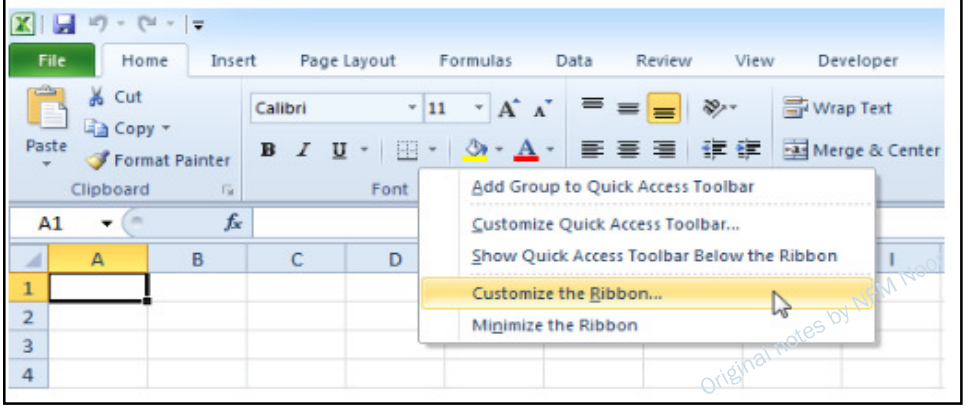
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Creating Macro

Developer Tab

To turn on the Developter tab, execute the following steps.

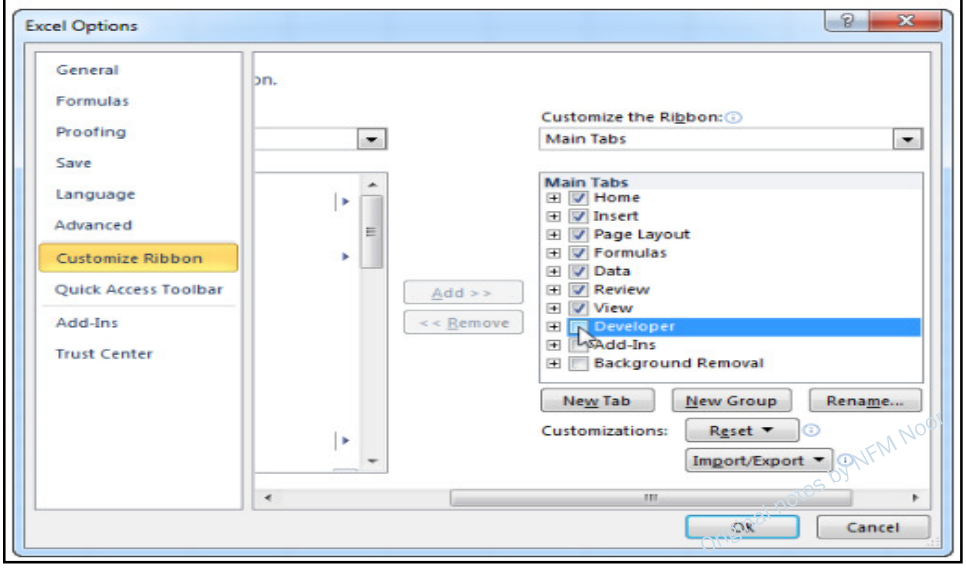
1. Right click anywhere on the ribbon, and then click Customize the Ribbon.



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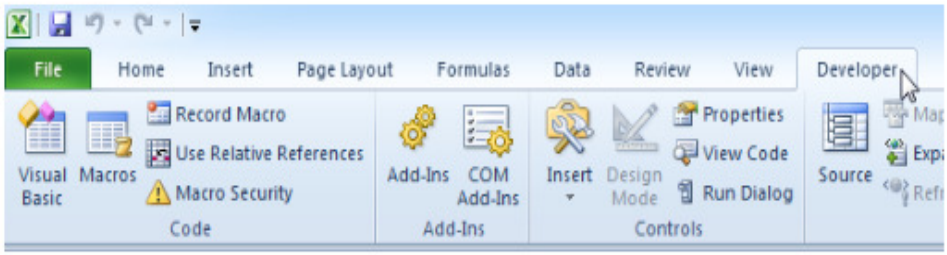
2. Under Customize the Ribbon, on the right side of the dialog box, select Main tabs (if necessary).

3. Check the Developer check box.



4. Click OK.

5. You can find the Developer tab next to the View tab.



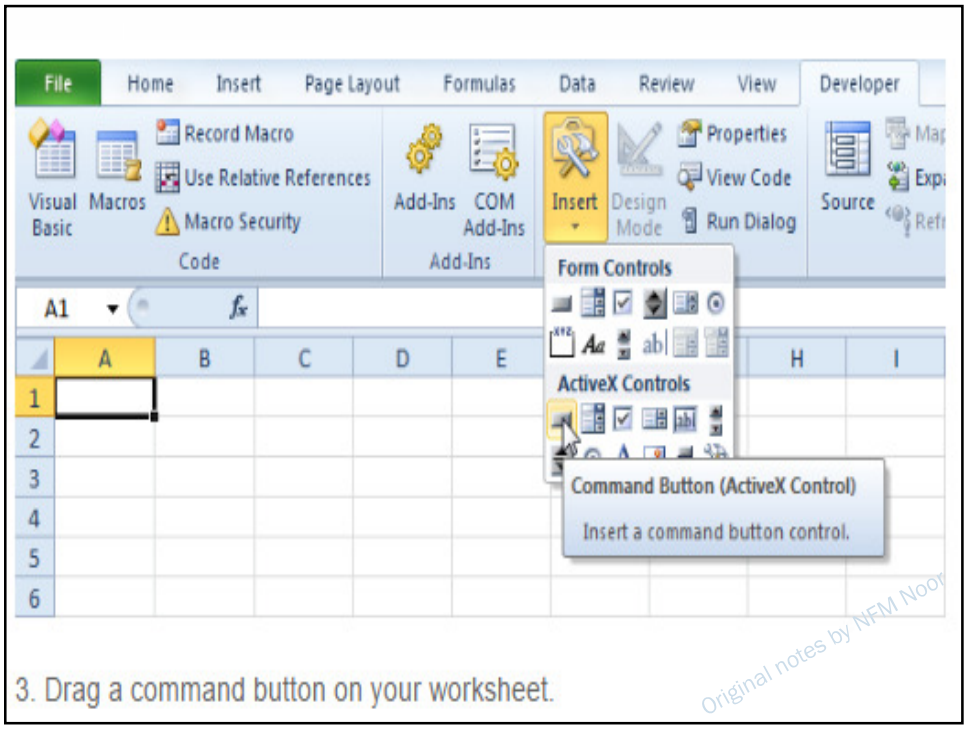
Command Button

To place a command button on your worksheet, execute the following steps.

1. On the Developer tab, click Insert.

2. In the ActiveX Controls group, click Command Button.

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3. Drag a command button on your worksheet.

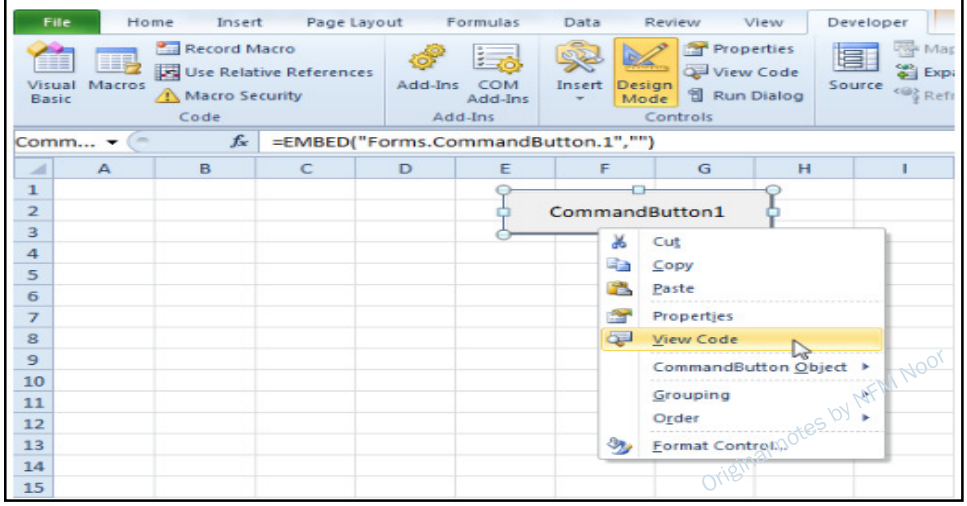
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Assign a Macro

To assign a macro (one or more code lines) to the command button, execute the following steps.

1. Right click CommandButton1 (make sure Design Mode is selected).

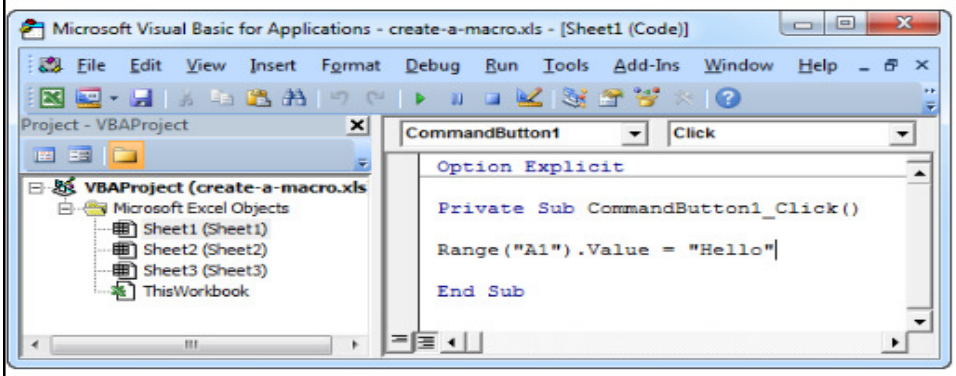
2. Click View Code.



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3. Place your cursor between Private Sub CommandButton1_Click() and End Sub.

4. Add the code line shown below.



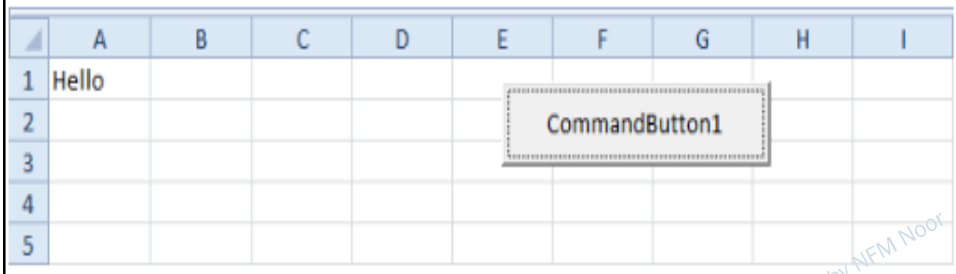
Note: the window on the left with the names Sheet1, Sheet2 and Sheet3 is called the Project Explorer. If the Project Explorer is not visible, click View, Project Explorer. To add the Code window for the first sheet, click Sheet1 (Sheet1).

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5. Close the Visual Basic Editor.

6. Click the command button on the sheet (make sure Design Mode is deselected).

Result:

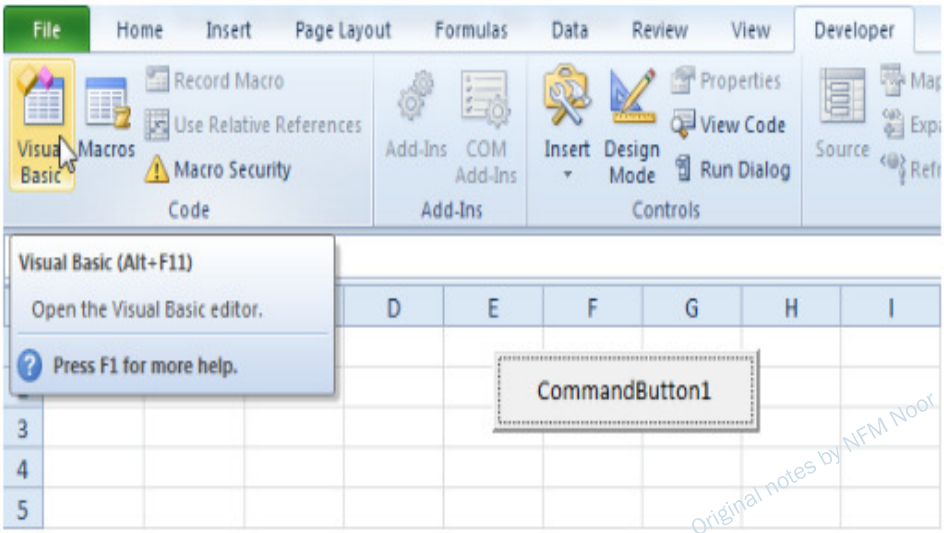


Congratulations. You've just created a macro in Excel!

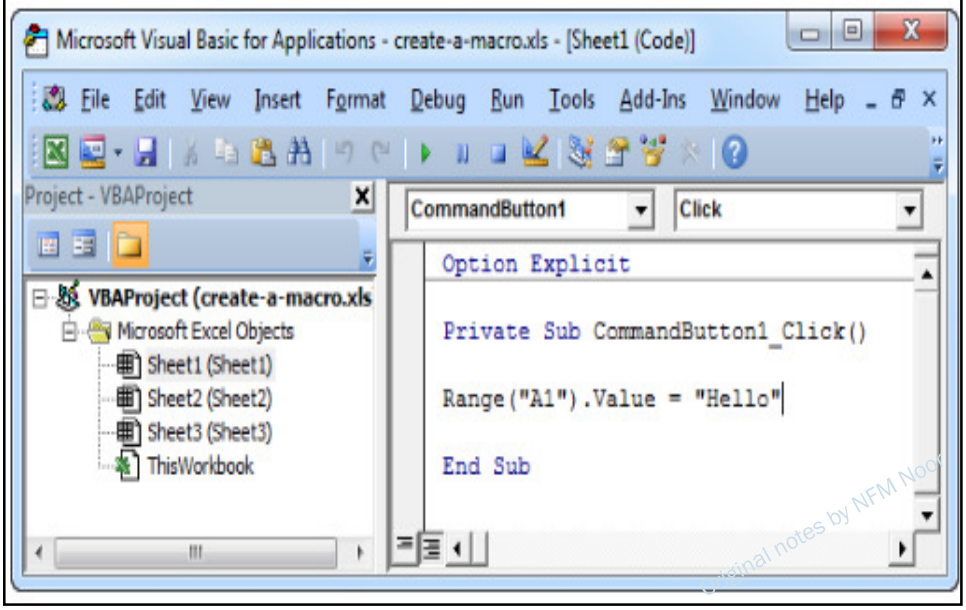
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Visual Basic Editor

To open the Visual Basic Editor, on the Developer tab, click Visual Basic.



The Visual Basic Editor appears.



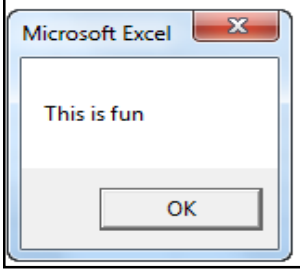
MsgBox

The MsgBox is a dialog box in Excel VBA you can use to inform the users of your program. Place a command button on your worksheet and add the following code lines:

1. A simple message.

```
MsgBox "This is fun"
```

Result when you click the command button on the sheet:



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Workbook and Worksheet Object

Object Hierarchy

In Excel VBA, an object can contain another object, and that object can contain another object, etc. In other words, Excel VBA programming involves working with an object hierarchy. This probably sounds quite confusing, but we will make it clear.

The mother of all objects is Excel itself. We call it the Application object. The application object contains other objects. For example, the Workbook object (Excel file). This can be any workbook you have created. The Workbook object contains other objects, such as the Worksheet object. The Worksheet object contains other objects, such as the Range object.

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The Create a Macro chapter illustrates how to run code by clicking on a command button. We used the following code line:

```
Range("A1").Value = "Hello"
```

but what we really meant was:

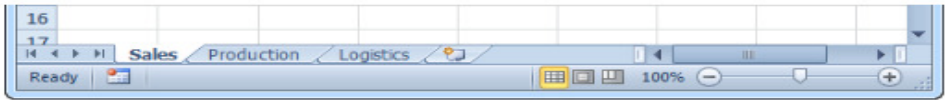
```
Application.Workbooks("create-a-macro").Worksheets(1).Range("A1").Value
= "Hello"
```

Note: the objects are connected with a dot. Fortunately, we do not have to add a code line this way. That is because we placed our command button in create-a-macro.xls, on the first worksheet. Be aware that if you want to change things on different worksheets, you have to include the Worksheet object. Read on.

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Collections

You may have noticed that Workbooks and Worksheets are both plural. That is because they are collections. The Workbooks collection contains all the Workbook objects that are currently open. The Worksheets collection contains all the Worksheet objects in a workbook.



You can refer to a member of the collection, for example, a single Worksheet object, in three ways.

1. Using the worksheet name.

```
Worksheets("Sales").Range("A1").Value = "Hello"
```

2. Using the index number (1 is the first worksheet starting from the left).

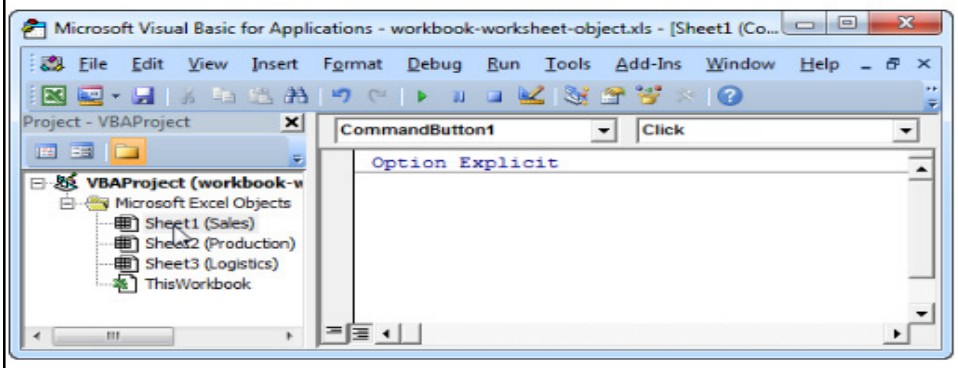
```
Worksheets(1).Range("A1").Value = "Hello"
```

3. Using the CodeName.

```
Sheet1.Range("A1").Value = "Hello"
```

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To see the CodeName of a worksheet, open the Visual Basic Editor. In the Project Explorer, the first name is the CodeName. The second name is the worksheet name (Sales).



Note: the CodeName remains the same if you change the worksheet name or the order of your worksheets so this is the safest way to reference a worksheet. Click View, Properties Window to change the CodeName of a worksheet. There is one disadvantage, you cannot use the CodeName if you reference a worksheet in a different workbook.

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Properties and Methods

Now let's take a look at some properties and methods of the Workbooks and Worksheets collection. Properties are something which an collection has (they describe the collection), while methods do something (they perform an action with an collection).

Place a command button on your worksheet and add the code lines:

1. The Add method of the Workbooks collection creates a new workbook.

```
Workbooks.Add
```

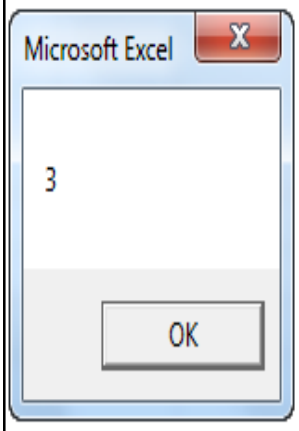
Note: the Add method of the Worksheets collection creates a new worksheet.

2. The Count property of the Worksheets collection counts the number of worksheets in a workbook.

```
MsgBox Worksheets.Count
```

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Result when you click the command button on the sheet:



Note: the Count property of the Workbooks collection counts the number of active workbooks.

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Range Object

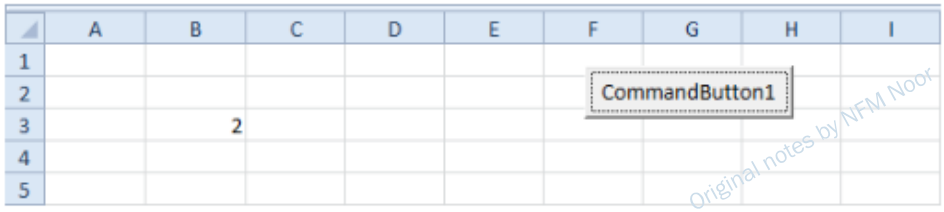
- Representation of cell (or cells) in worksheet.

Range Examples

Place a command button on your worksheet and add the following code line:

```
Range("B3").Value = 2
```

Result when you click the command button on the sheet:

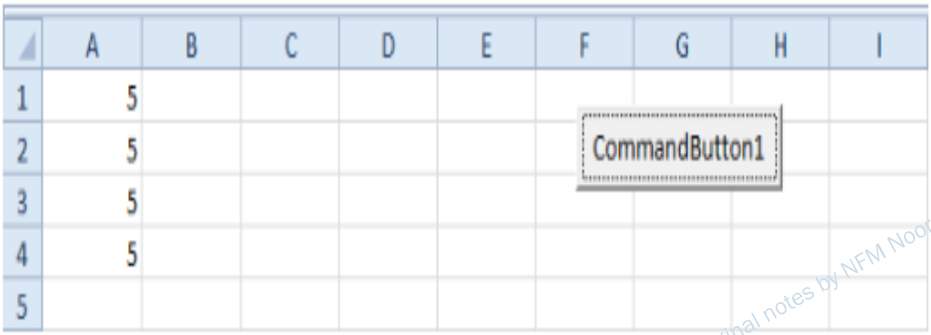


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Code:

```
Range("A1:A4").Value = 5
```

Result:

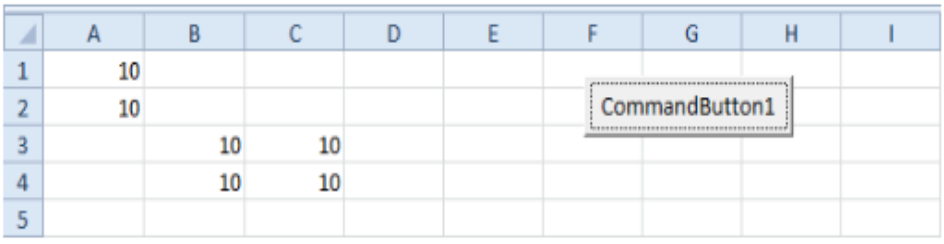


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Code:

```
Range("A1:A2,B3:C4").Value = 10
```

Result:



Note: to refer to a named range in your Excel VBA code, use a code line like this:

```
Range("Prices").Value = 15
```

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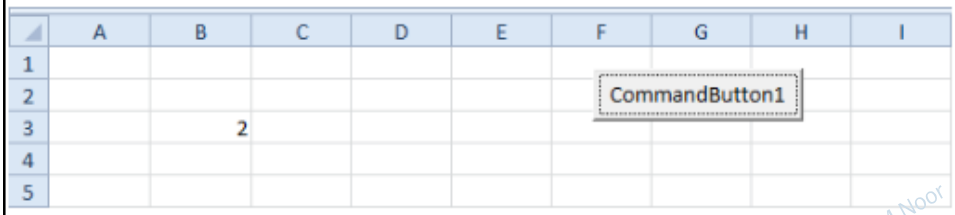
Cells

Instead of Range, you can also use Cells. Using Cells is particularly useful when you want to loop through ranges.

Code:

```
Cells(3, 2).Value = 2
```

Result:



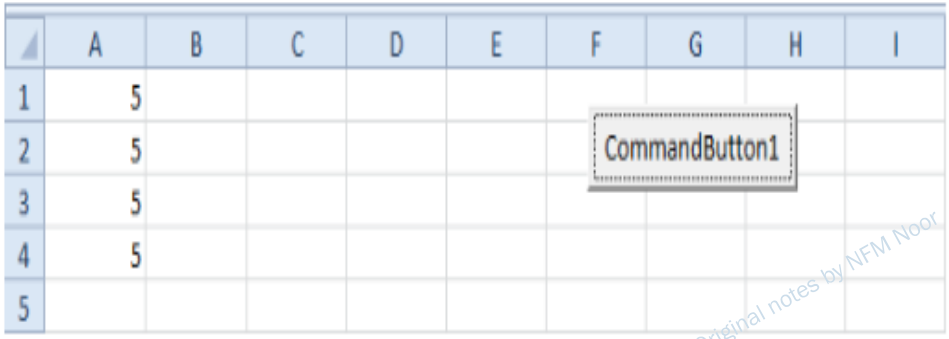
Explanation: Excel VBA enters the value 2 into the cell at the intersection of row 3 and column 2.

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Code:

```
Range(Cells(1, 1), Cells(4, 1)).Value = 5
```

Result:



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Declare a Range Object

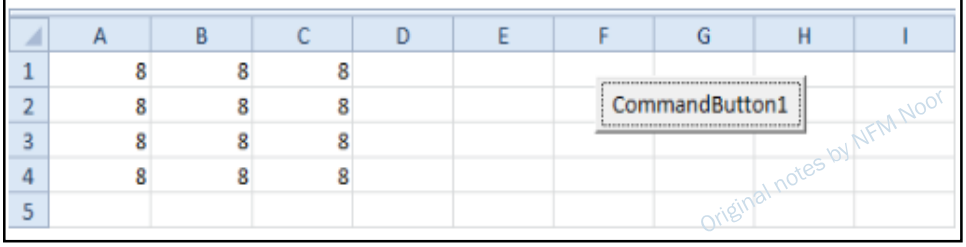
You can declare a Range object by using the keywords Dim and Set.

Code:

```
Dim example As Range
Set example = Range("A1:C4")

example.Value = 8
```

Result:



Select

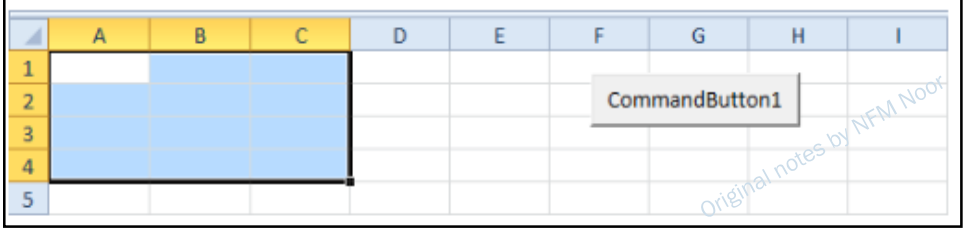
An important method of the Range object is the Select method. The Select method simply selects a range.

Code:

```
Dim example As Range
Set example = Range("A1:C4")

example.Select
```

Result:



Rows

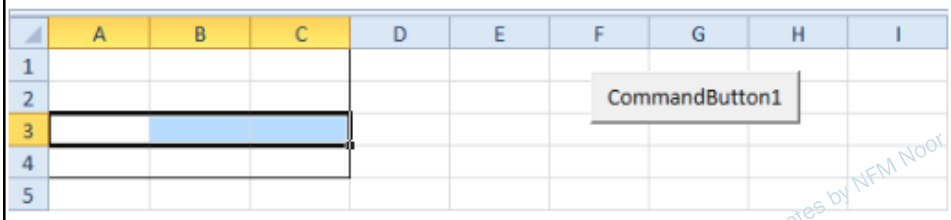
The Rows property gives access to a specific row of a range.

Code:

```
Dim example As Range
Set example = Range("A1:C4")

example.Rows(3).Select
```

Result:



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Note: border for illustration only.

Columns

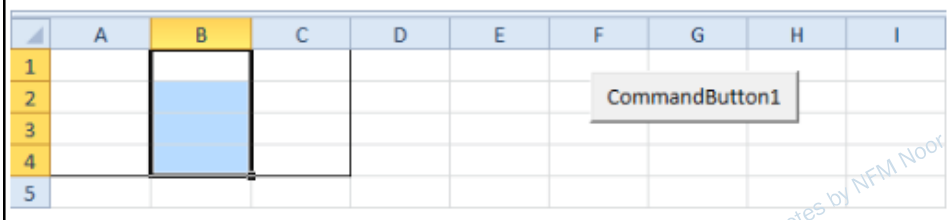
The Columns property gives access to a specific column of a range.

Code:

```
Dim example As Range
Set example = Range("A1:C4")

example.Columns(2).Select
```

Result:



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Note: border for illustration only.

Copy/Paste

The Copy and Paste method are used to copy a range and to paste it somewhere else on the worksheet.

Code:

```
Range("A1:A2").Select
Selection.Copy

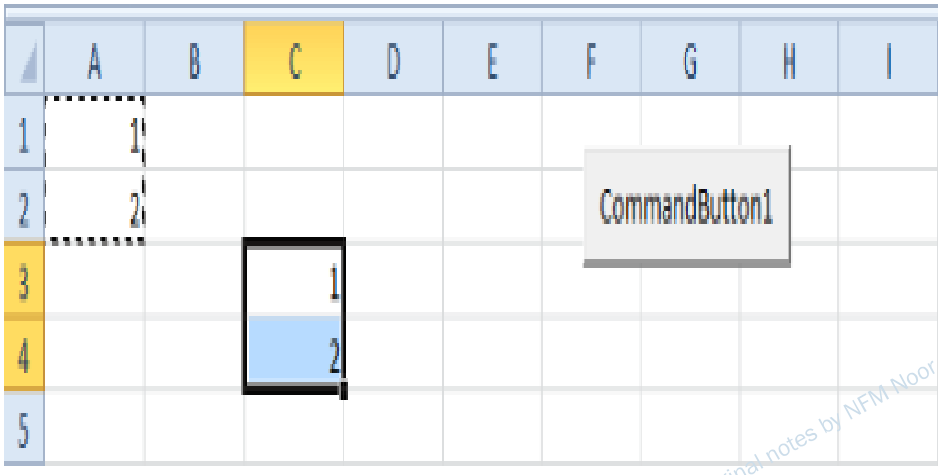
Range("C3").Select
ActiveSheet.Paste
```

Although this is allowed in Excel VBA, it is much better to use the code line below which does exactly the same.

```
Range("C3:C4").Value = Range("A1:A2").Value
```

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Result:



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Clear

To clear the content of an Excel range, you can use the ClearContents method.

```
Range("A1").ClearContents
```

or simply use:

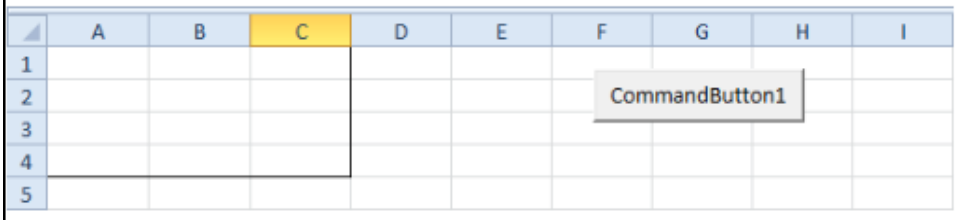
```
Range("A1").Value = ""
```

Note: use the Clear method to clear the content and format of a range. Use the ClearFormats method to clear the format only.

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Count

With the Count property, you can count the number of cells, rows and columns of a range.



Note: border for illustration only.

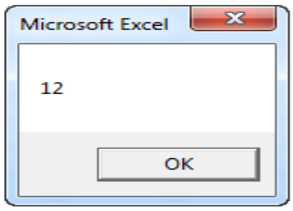
Code:

```
Dim example As Range
Set example = Range("A1:C4")

MsgBox example.Count
```

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Result:

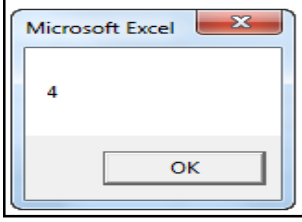


Code:

```
Dim example As Range
Set example = Range("A1:C4")

MsgBox example.Rows.Count
```

Result:



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Variables

- Declaring a variable is to introduce that variable to Excel VBA.
- Initializing a variable is to assign initial value to a variable.
- Types of Variables: Integer, Double, String, Boolean

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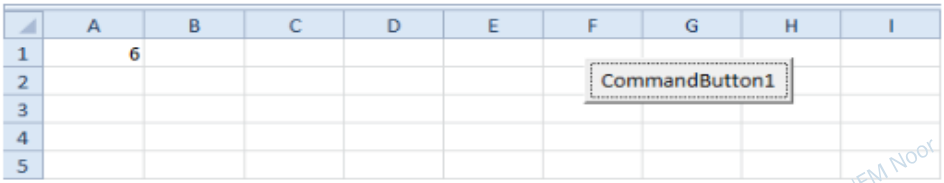
Place a command button on your worksheet and add the code lines below. To execute the code lines, click the command button on the sheet.

Integer

Integer variables are used to store whole numbers.

```
Dim x As Integer
x = 6
Range("A1").Value = x
```

Result:



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Explanation: the first code line declares a variable with name x of type Integer. Next, we initialize x with value 6. Finally, we write the value of x to cell A1.

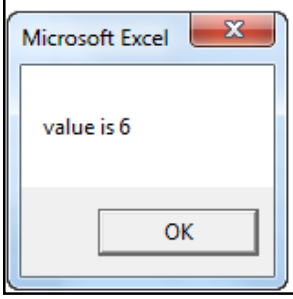
Double

A variable of type Double is more accurate than a variable of type Integer and can also store numbers after the comma.

Code:

```
Dim x As Integer
x = 5.5
MsgBox "value is " & x
```

Result:



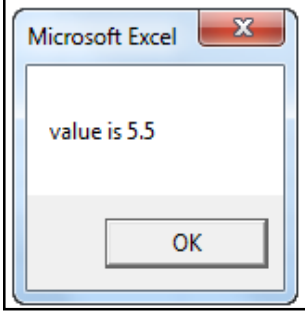
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But that is not the right value! We initialized the variable with value 5.5 and we get the value 6. What we need is a variable of type Double.

Code:

```
Dim x As Double
x = 5.5
MsgBox "value is " & x
```

Result:



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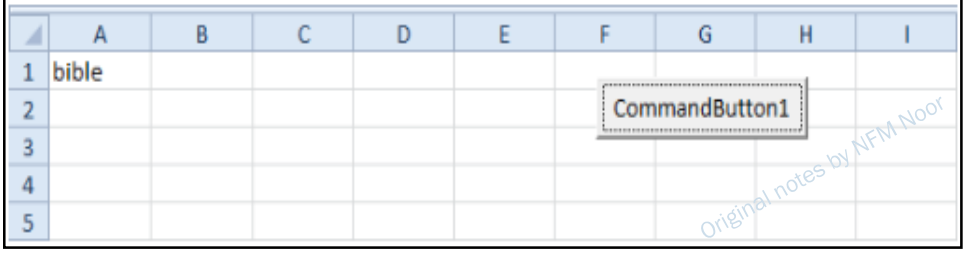
String

String variables are used to store text.

Code:

```
Dim book As String
book = "bible"
Range("A1").Value = book
```

Result:



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Boolean

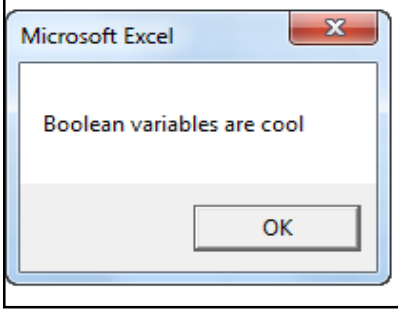
Use a Boolean variable to hold the value True or False.

Code:

```
Dim continue As Boolean
continue = True

If continue = True Then MsgBox "Boolean variables are cool"
```

Result:



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Summary

- Excel provides tools for basic programming with VBA.
- CommandButton is a basic VBA that can easily be assigned with a Macro.
- Understanding workbook and worksheet objects is a key of a successful VBA project.
- Integer, Double, String, Boolean are variables that can be operated and manipulated in VBA.

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